



LIFESYNC



Autonomous Emergency Connectivity



Problem Statement Title

**Emergency Care
System – Real-Time
Ambulance
Connectivity**



Theme / Category

**MedTech /
Healthcare
(Software)**



Team Name

Delta Devs

THE PROBLEM

The Information Blindspot in Emergency Transit

Disconnected Systems

Information Blindspot:

Hospitals lack access to live patient vitals and medical history during transit.

Delayed Preparation:

Doctors cannot prepare trauma teams or theaters without real-time GPS and monitoring.

Fragmentation:

Responders operate on systems with minimal interoperability.

The Critical Window

The transit time represents a **critical window** for lifesaving intervention.

Current communication gaps restrict potential medical actions, turning ambulances into isolated "black boxes" of data.





LIFESYNC – Autonomous Emergency Connectivity

PROPOSED SOLUTION

Breaking the "Information Blindspot"

A unified platform streaming patient vitals, GPS, and live video from ambulances to hospital dashboards. Doctors can prepare theaters or guide paramedics remotely.



Smart Auto-Routing Agentic AI

Automatically reroutes to hospitals with available specialists based on live patient vitals.



Early Warning System Predictive AI

Analyzes trends to calculate "Risk Scores," alerting doctors before a patient crashes.



Connectivity "Dead Zone" Offline Mode

Caches data during signal loss and "burst-syncs" once connection is restored.



Security Tamper-Proof Diary

SHA-256 cryptographic hashing ensures medical event logs are mathematically secure.



CORE FEATURES

The LifeSync Integrated Emergency System



Live Vitals & GPS Streaming

Continuously streams real-time ECG, SpO2, and HR with precision GPS tracking to hospital dashboards.



Two-Way Remote Guidance

Seamless video/audio allows specialists to visually assess patients and guide paramedics in real-time.



Predictive Risk Alerts (AI)

Background AI analyzes trends to calculate a "Risk Score," alerting teams before patient deterioration.



Autonomous Rerouting

Agentic AI reroutes to facilities with available specialists based on specific patient trauma needs.



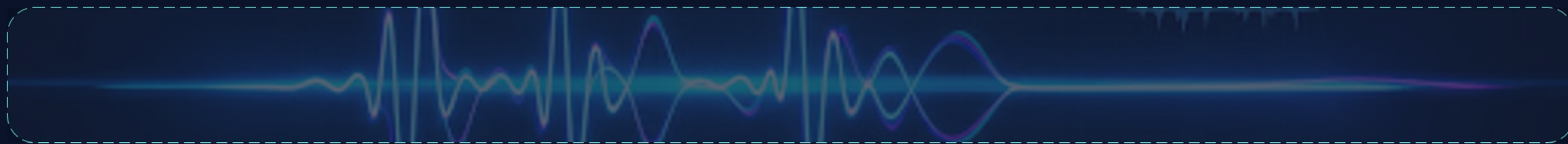
"Dead Zone" Resilience

Local caching ensures no data loss; "burst-syncs" missing vitals the moment connection is restored.



Tamper-Proof Ledger

SHA-256 cryptographic hashing creates a legally sound, unalterable timeline of the medical response.



HOW IT WORKS

The LifeSync Data Pipeline

01



Capture & Secure

Paramedics log details via React Native app to capture vitals and GPS.

Security Layer:

SHA-256 hash locks data into an immutable "Chain of Custody."

02



Stream & Buffer

Instant push via Node.js/WebSockets for zero-latency streaming.

Resilience Layer:

Offline "Dead Zone Mode" caches vitals for burst-sync upon signal return.

03



Intelligent Analysis

Python microservices analyze trends to calculate live Risk Scores.

Agentic AI:

Checks trauma needs to automate rerouting to specialized facilities.

04



Proactive Intervention

React.js dashboard displays live maps, vitals, and WebRTC video.

Trauma Alert:

Visual alerts trigger ER prep before the ambulance arrives.

TECHNICAL APPROACH



Methodology

- **Data Collection:** Paramedic app collects, encrypts, and caches patient vitals using React Native.
- **Instant Sync:** Node.js backend streams data in real-time via Socket.io for zero latency.
- **AI Analysis:** Python microservice analyzes streams for risk alerts and "Deterioration Index".
- **Security:** SHA-256 hashing ensures a tamper-proof medical "Chain of Custody".
- **Consultation:** Low-latency two-way emergency calls via WebRTC.

Core Technologies

React Native: Cross-platform mobile deployment.

FastAPI & Scikit-Learn: High-performance AI microservices.

Leaflet.js: Free, open-source GPS mapping integration.

React.js & Tailwind: High-contrast, dark-mode hospital UI.

WebRTC: Peer-to-peer real-time video communication.



FEASIBILITY & STRATEGIC SOLUTIONS

Economic Viability

Zero Hardware Dependency

Leverages existing paramedic smartphones/tablets.

Low Operational Cost

Built on 100% free, open-source tech stacks.

High Scalability

Node.js handles thousands of ambulance connections.

Challenges & Risks

The "Dead Zone"

Intermittent connectivity in rural or highway areas.

Doctor "Alert Fatigue"

Constant dashboard flashes are ignored by busy ER staff.

Data Integrity & Trust

Legal requirement to prove vitals weren't altered.

Solutions & Risk Mitigation

"Store & Forward" Offline Burst

AsyncStorage caches data during drops; burst-syncs or falls back to compressed SMS upon restoration.

AI Deterioration Index

Python AI calculates holistic "Risk Trends." Dashboard alerts only trigger when a crash is predicted.

Cryptographic Chain of Custody

SHA-256 hashing "locks" every payload, allowing hospitals to mathematically verify 100% accuracy.



Built with 100% Free/Open-Source tools (OpenStreetMap, Leaflet.js) for zero cost.

IMPACT & BENEFITS

POTENTIAL IMPACT ON TARGET AUDIENCE

Patients (The "Golden Hour" Advantage)

Shifts emergency care from **reactive** to **proactive**. Significantly reduces mortality rates by ensuring specialists and equipment are waiting at the door the exact second the stretcher arrives.

Paramedics (Cognitive Relief & Empowerment)

Reduces on-field panic. Turns isolated ambulance crews into fully supported extensions of the hospital via live WebRTC guidance and AI background assistance, allowing them to focus purely on the patient.

BENEFITS OF THE SOLUTION

Economic (Hospital Resource Optimization)

Eliminates costly "standby waste" and ER bottlenecks. Hospitals can precisely allocate ICU beds, blood banks, and trauma surgeons based on verified live data, not just radio guesswork.

Social & National Scale

Directly aligns with National Health Mission (NHM) objectives. It transforms a fragmented, blind transit system into a unified, interoperable healthcare grid deployed across urban centers and rural "dead zones."



PHASE 2.0: THE FUTURE ROADMAP

Scaling LifeSync into a National Emergency Grid



Smart City: Traffic Preemption

Integration: Linking LifeSync GPS Streaming API with municipal Traffic Management Systems.

Outcome: Real-time "Green Wave" activation, reducing transit time by **25–30%** in urban areas.



ER Supply-Chain Auto-Trigger

Logic: Agentic AI analyzes live vitals (blood loss/trauma type) to alert pharmacy and blood banks.

Outcome: Essential blood and meds are prepped and waiting at the ER bay before arrival.



Mass Casualty "Triage Mode"

Scaling: Expanding Predictive AI Risk Score to handle 50+ simultaneous ambulance connections.

Outcome: Centralized command prioritizing beds based on the "Deterioration Index."

Aligning Software-based MedTech with National Health Mission (NHM) Interoperability Objectives

RESEARCH & REFERENCES



National Health Mission (NHM)

Guidelines on emergency infrastructure integration and ambulance connectivity gaps in India's digital health ecosystem.



WebRTC Technical Standards

Research on deploying low-latency video and audio streaming for telemedicine in low-bandwidth, high-mobility environments.



Clinical Deterioration Studies

Medical research on Early Warning Scores (EWS) to model AI Risk logic for predicting patient emergencies based on Vitals trends.



A step toward India's digital health transformation.